Aran Handoff

Aran: handoff pack

Prepared 7 Nov 2026 by Inbathamizhan S so another person (and their Claude) can pick up Aran exactly where it stands. Read this file first, then `repo/README.md`.

1. What Aran is

An algebraic data-flow firewall for Indian banks. It finds Aadhaar, PAN, card numbers, UPI IDs and OTPs in APIs, logs and AI tools, replaces them with tokens before they are stored or shared, traces any leak back to the system that leaked it, and supports DPDP-compliant erasure with signed receipts.

The core trick: real Aadhaar and card numbers always pass their check digit (Verhoeff, Luhn); Aran's tokens are built to always fail it, so any scanner can prove whether a number is protected or leaked. On 32,000 real Loghub log lines, false alarms fell from 67,481 to 6 (OTP), 6,086 to 81 (card) and 366 to 4 (Aadhaar), with 0 of 20,000 tokens wrongly flagged.

Built for the BuildX hackathon, team Iravora Void, team ID BX-062. Domain: Banking, Financial Services and FinTech.

2. Team (KARE, Kalasalingam Academy of Research and Education)

Person	Role
Inbathamizhan S	Lead, cybersecurity (GitHub `itzz-inbathamizhanS`)
Aasif Ahmed A	Cybersecurity
SN Arun Vijay	AI/ML (GitHub `ArunvijaySNAV`)
Bharadhwaj M	AI/ML

Everyone is new to this tech, which is why the repo has 30+ beginner learning docs.

3. Rules every new Claude session must follow

1. On every prompt, first run the `model-router` skill (pick the cheapest fitting model: haiku, sonnet or opus) and follow it. The other user needs this skill installed in their own Claude account; its text is in `rules/model-router-skill.md`.
2. Before any new plan, first write beginner learning docs (one per topic) for the team.
3. Every artifact or update is saved into the repo under `artifacts/<topic>/` (Markdown and PDF), the main `README.md` gets links to it, and it is pushed.
4. All commits are authored as **Inbathamizhan S**. No Claude or AI-assistant words anywhere: no Co-Authored-By lines, no session links, no "generated by" lines.

5. Never use a real person's Aadhaar, PAN or card. Test values are generated with valid checksums.
6. Change `main` through a pull request. Move old things to `archive/` rather than deleting them.
7. Obsidian habit (from `notes/Workflow.md`): log each change in `notes/daily/<date>.md`, add new branches to `notes/Branches.md`, a note per thread in `notes/threads/`.

A ready-to-paste instruction block for a new Claude project is in `rules/PROJECT-INSTRUCTIONS.txt`.

4. Where the code is

- GitHub: https://github.com/itzz-inbathamizhanS/Aran (the owner has to add the new person as a collaborator).
- Local working copy on the lead's PC: `D:\Aran` (also the Obsidian vault).
- `repo/` in this pack is a snapshot of `main` at commit `6f8e633` ("Aran BuildX final submission (BX062)"), without the videos.

Branch	State
`main`	Current. Everything live runs from here and redeploys on every push.
`FaaS`	Firewall as a service. Merged into main (PR #18, #19).
`BanksAndCloud`	Arun Vijay's first Aran Cloud + demo bank work. Audited 31 Oct and not merged as-is (it committed a .venv and databases); Aran Cloud and the banks now live on main. Audit: `repo/artifacts/Branch audits/`.
`aran-ui-banks-cloud`	One UI for console, cloud and banks (Learn 23). Check `git log main..aran-ui-banks-cloud` before deleting it.
`Pre_copy_of_main`	Backup of main before the 4 Nov cleanup. No longer on GitHub as of 7 Nov; `archive/` on main holds the moved files.

No pull requests are open.

5. Live links

What	Link
Firewall console (main demo)	https://aran-console.vercel.app
Console, direct on Render	https://aran-console.onrender.com
Aran Cloud (SaaS, firewall as a service)	https://aran-cloud.vercel.app
Bank A (legacy, stores raw identifiers)	https://aran-bank-a.vercel.app
Bank B (protected by Aran)	https://aran-bank-b.vercel.app

Render sleeps when idle, so the first visit can take about a minute. A GitHub Actions job (`.github/workflows/keep-alive.yml`) pings it every 5 minutes, 06:30 to 23:30 IST, until 19 Nov 2026. Vercel and Render accounts belong to the lead; the new person needs access to redeploy.

6. What is in this pack

Folder	Contents
`HANDOFF.md`	This file
`rules/`	Project instructions and the model-router skill text
`repo/`	Snapshot of `main`: code (`app/`, `aran_cloud/`, `aran_edge/`, `aran_sdk/`, `demo_bank/`, `research/`), docs, all artifacts except videos
`repo/docs/learning/`	Beginner learning docs, Learn 1 to 32
`repo/docs/research/`	Research doc, AI and immune layers, roadmap, normaliser, ledger and gateway, post-quantum
`repo/artifacts/`	Every write-up and PDF: Project Story, Pitch deck, BuildX submission (BMC + deck), audits, firewall as a service, UI work, screenshots
`obsidian-vault/`	The vault on its own: `Home.md`, `notes/` (Workflow, Branches, Threads, Artifacts, daily notes, 20 thread notes), `templates/`, `.obsidian/` settings
`notion-handoffs/`	Copies of the Notion “Second Brain / Projects / Aran” handoff pages (15 to 16 Oct 2026)

The best single read is `repo/artifacts/Project Story/Aran - Full Project Story from Zero.md` (every term explained, from zero to now).

Videos are not in the zip (about 340 MB). They are in the repo under `artifacts/Explainer video/` (24-minute quick tour) and `artifacts/Explainer video v2/` (76-minute course in 6 parts).

There were no separate Notepad (.txt) notes in the repo; the running notes live in the Obsidian vault and Notion, both included.

7. Open items

- Fix the two detector regressions on unseen phrasings: an OTP value before “PIN”, and “cheque no” taken as a mobile number. Short cues without a colon (“mob 98…”) are still missed.
- Names are the weakest detector: add transformer NER behind the gate (roadmap item 7), Tamil NER (item 8), OCR on KYC images (item 10), immune layer on log templates (item 11), vectorised speed-up (item 12).
- Run the AI gateway against a real model.
- Hybrid post-quantum TLS (X25519MLKEM768) in front of the proxy and AI gateway; SLH-DSA on Merkle roots.
- Get a legal view on crypto-shredding as DPDP erasure (roadmap item 13).
- Set the GitHub repo description.
- Keep-alive job ends 19 Nov 2026; extend it if the demo is still needed.
- Delete merged feature branches when convenient.

Full roadmap: `repo/docs/research/aran-roadmap.md`.

8. How to run it locally

```
git clone https://github.com/itzz-inbathamizhanS/Aran
cd Aran
docker compose up --build  # console :8000, Aran Cloud :8100, Bank A :9001,
Bank B :9002
```

Without Docker: make a venv, `pip install -r app/server/requirements.txt`, then `./scripts/run_demo.sh`. Core and console only: `cd app/server && uvicorn main:app --port 8000`. Windows setup with fixes for common errors: `repo/docs/learning/01-python-and-tools.md`. More options in `repo/README.md` under “Run it”.

9. First prompt for the new Claude

> Read HANDOFF.md, then README.md and notes/Threads.md in the Aran repo. Follow the rules in section 3. Summarise where the project stands and propose the next step from the open items, starting with beginner learning docs for any new topic.